Attention Is All You Need - An Overview

The Transformer is a neural network architecture based entirely on attention mechanisms, dispensing with recurrence and convolutions entirely. The dominant sequence transduction models before the Transformer were based on complex recurrent or convolutional neural networks that include an encoder and a decoder.

The attention mechanism can be described as mapping a query and a set of key-value pairs to an output, where the query, keys, values, and output are all vectors. The output is computed as a weighted sum of the values, where the weight assigned to each value is computed by a compatibility function of the query with the corresponding key.

Scaled Dot-Product Attention

The particular attention used in the Transformer is called Scaled Dot-Product Attention. The input consists of queries and keys of dimension dk, and values of dimension dv. We compute the dot products of the query with all keys, divide each by the square root of dk, and apply a softmax function to obtain the weights on the values.

Multi-Head Attention allows the model to jointly attend to information from different representation subspaces at different positions. Instead of performing a single attention function, the queries, keys and values are linearly projected h times with different learned projections, attention is applied in parallel, and the outputs are concatenated.

Why Self-Attention

Self-attention, sometimes called intra-attention, is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention layers are faster than recurrent layers when the sequence length is smaller than the representation dimensionality, and they allow significantly more parallelization.

Positional encodings are added to the input embeddings to inject information about the relative or absolute position of tokens in the sequence, since the model contains no recurrence and no convolution.